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Life cycle assessment of binary recycled ceramic tile and recycled brick waste-based geopolymers

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ABSTRACT

This paper consists of a life cycle assessment (LCA) analysis on an experimental study to produce recycled brick waste (RBW) and recycled ceramic tile (RCT) based geopolymers. The LCA analysis is done following ISO 14040 standards on GaBi software for geopolymers produced with different ingredients ratios and curing types/regimes. The binary geopolymer comes from RBW + RCT (Phase I) as aluminosilicate sources and alkaline solutions. The ternary geopolymer (Phase II) studied includes Class C Fly ash (FC) in addition to the RBW + RCT. And finally, for Phase III, the binary geopolymer is cured at 100 °C. The life cycle impact assessment (LCIA) methodology using TRACI 2.0 and the GaBi databases is used to calculate the following impact categories: Global Warming Potential (GWP), Acidification Potential (AP), Eutrophication Potential (EP), Ozone Depletion Potential (ODP) and smog. The LCA analysis is also modelled on a large-scale industrial basis. This results in decreased environmental impacts, and they are 1352 kg CO_2-eq, 3.01 kg SO_2-eq, 0.19 kg N-eq, 6.56E-9 kg CFC 1-eq for GWP, AP, EP, and ODP, respectively.

Introduction

The rate of urbanisation is increasing globally, and it is predicted that up to 68% of the entire population will live in urban areas by 2050 (United Nations Department of Economic, 2018). Concrete is one of the most widely used substances on the earth, and its use is growing exponentially. The Portland cement (PC), the fundamental component of concrete, and the production of this material contribute significant greenhouse gas (GHG) emissions. PC contributes around 8–9% to the global carbon dioxide (CO_2) emissions (Monteiro et al., 2017; Khan et al., 2021). These emissions are due to the energy resources required during production and the calcination process to convert limestone to lime and CO_2. Clinker manufacturing, a kilned and quenched material used in PC production, contribute most to the emissions (Andrew, 2018).

Construction and demolition waste (CDW) consists of concrete, clay brick, ceramic, mortar, wood, plastic, and steel. The USA produced 534 million tons of CDW in 2014, and the EU had 855 million tons per year (Silva et al., 2019). As urbanisation increases, there is a need to recycle this enormous amount of waste materials and to find an environmentally friendly solution to prevent damage to future generations. It can be achieved by recycling CDW by producing "green" construction materials in the construction sector (Zanelli et al., 2021). This practice can reduce environmental and economic footprints (Liu et al., 2014). Brick and ceramic tile waste are produced at the construction sector's demolition, construction, sintering, and transportation stages, making up to 50% of the total CDW globally (Silva et al., 2019; Liu et al., 2014; Huynh et al., 2021). There is also a growing global use of ceramic tiles and clay bricks, especially in developing countries and China. This leads to large amounts of brick and ceramic tile waste, especially fine particles smaller than 150 μm, making up to 20% of the CDW (Mahmoodi et al., 2021).

Geopolymers, which are inorganic polymer binders, are formed by the alkaline activation of amorphous aluminosilicate-based powder materials (e.g., blast furnace slag, fly ash, calcined clays, natural pozzolans) with alkali-based solutions (e.g., sodium hydroxide, potassium hydroxide, calcium hydroxide, sodium silicate) (Duxson et al., 2007; Hamdi et al., 2019; Guo et al., 2010; Provis, 2018; Shi et al., 2019; Xu et al., 2018). Colangelo et al. (Colangelo et al., 2021) found that eco-sustainable geopolymer binders have advantages that make them a good alternative to traditional binders. They can be used as alternatives

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Nomenclature

AP	Acidification Potential
CDW	Construction and Demolition Waste
CO_2	Carbon Dioxide
EP	Eutrophication Potential
FC	Fly ash – C class
FF	Fly ash – F class
GGBS	Ground Granulated Blast Furnace Slag
GWP	Global Warming Potential
LCA	Life Cycle Assessment
LCI	Life-Cycle Inventory
LCIA	Life Cycle Impact Assessment
M	Metakaolin
SS	Sodium Silicate
SCS	Supplementary Cementitious Materials
ODP	Ozone Depletion Potential
OPC	Ordinary Portland Cement
PC	Portland Cement
RBW	Recycled Brick Waste
RCT	Recycled Ceramic Tile

to PC and minimise CO_2 emissions. Several studies have analysed different source materials, alkali solutions, and optimisation strategies to produce geopolymer binders with high durability and mechanical properties. The mainstream aluminosilicate precursors mentioned above have some drawbacks that may limit their widespread use in geopolymer binder production, such as non-availability in all locations, high transportation costs, and relatively high costs due to high demand. (Guo et al., 2010; Ulugöl et al., 2021). Considering all these, alternative source materials (i.e., precursors) are required for producing geopolymers more effectively and innovatively. CDWs (e.g., concrete, bricks, tiles, ceramics, and glass) are potential candidates to be used as aluminosilicate sources for geopolymerization. Luhar et al. (Luhar et al., 2021) reviewed the use of ceramic waste in geopolymer composites production and found that it lowered the energy requirements and costs of materials in the construction sector. Several other studies in the current literature focused on using CDW-based materials in geopolymer production (Ulugöl et al., 2021; Dadsetan et al., 2019), and the results of these studies are very promising.

Evaluating the environmental impacts of producing geopolymers alternatives to Portland cement is essential. It can be done using a life cycle assessment (LCA) analysis, a sustainable development tool used to measure and compare the environmental impacts of goods and services (Alhaj et al., 2022; Khan et al., 2022). These ecological impacts include climate change, stratospheric ozone depletion, smog creation, eutrophication, and acidification. The product or service can be analysed over its entire life cycle, including the design, production/manufacturing, use, and end-of-life phases (Rebitzer et al., 2004). Bajpai et al. (2020) performed an environmental impact assessment on fly ash-silica fume geopolymers. They found that the geopolymer concretes lowered the global warming potential (GWP) compared to cement-based ones. They used ReCiPe midpoint and endpoint indicators in the UMBERTO NXT software using Ecoinvent 3.0. Shi et al. (2021) found that the CO_2 emissions of the geopolymer binder studied were lower than that of ordinary PC at the same compressive strength. Moreover, at 70 MPa, the CO_2 emissions of the geopolymer were reduced by 166.36 kg $CO_2/m^{3,}$ which was 62.73% of its emissions. They also found that increasing the compressive strength of the geopolymer had no significant effect on CO_2 emissions. Finally, they concluded that the CO_2 emissions mainly depend on the alkali activator solution, transportation, and heat curing. Ghadir et al. (2021) studied volcanic ash-based geopolymers and performed a comparative LCA analysis with a functional unit of 1 m^3. They found that the geopolymers had a similar climate change impact to cement (CEM1); however, they concluded that their boundary conditions might be the main contributor to the environmental effects. However, some studies suggest that not all alkali-activated cement mixtures would have a lower GWP than cement. The future trends lead to a hybrid of geopolymer and cement-based concrete products (Ouellet-Plamondon and Habert, 2015).

In this study, an LCA analysis will be carried out on an experimental paper producing CDW-based geopolymer products as alternatives to the PC (Mahmoodi et al., 2021). The empirical research focuses on RBW and RCT-based geopolymers with varying compositions and initial treatment temperature conditions to find the optimum product based on compressive strength and physical properties. There is a need to carry out an environmental impact assessment of the developed material at an early stage to ensure the developed product's ecological sustainability and process. Therefore; a cradle-to-gate LCA analysis following ISO 14040 standards is conducted.

In this study, the objectives are listed as follows:

- Conduct a cradle-to-gate LCA analysis for RBW and RCT-based geopolymers following ISO 14,040 standards using GaBi software;
- Perform the life cycle impact assessment (LCIA) on Phase I, II, and III compositions in which Phase I includes a binary (RBW + RCT) geopolymer, Phase II consist of a ternary (RBW + RCT + supplementary cementitious materials (SCMs)) geopolymer. Phase III includes a binary (RBW + RCT) geopolymer and different curing temperatures.
- Perform an LCA case study where large-scale industrial machines are used for crushing and grinding the precursors instead of lab-scale ones.

System description

The LCA analysis is performed on a previous experimental paper (Mahmoodi et al., 2021), which studies the practical approach to combining RBW and RCT in binary geopolymer binders and RBW, RCT, and SCM in ternary geopolymer binders. The simplified flow chart for the procedure is shown in Fig. 1; however, the detailed version can be found in the original paper (Mahmoodi et al., 2021).

The RCT and RBW were obtained from a CDW sorting & management site. These two materials underwent a jaw crusher, milling, and sieving process to reduce the fragments' size and produce CDW-based precursor powders. After the crushing, milling, and sieving, CDW-based geopolymer mixtures are made using sodium hydroxide (SH) and sodium silicate (SS) as alkaline activators. In Phase I, the binary geopolymers was designed at 20 to 80% of RBW and RCT and pre-estimated SiO_2/Al_2O_3 and Na_2O/SiO_2 ratios of all composition. In Phase II, the effect of the incorporation of metakaolin (M), ground granulated blast furnace slag (GGBS), fly-ash-C class (FC), and fly ash-F class (FF) (at 15, 30, and 45% replacement rates) were investigated by designing ternary RBW + RCT + SCM mixtures with optimum binary combinations (from Phase I) under ambient temperature. In Phase III, to study the effect of curing temperature, the optimal medicines (from Phase I) were studied under high-temperature exposure (50, 75, and 100 °C) for 24 h after casting. A fixed liquid-to-solid (L/S) ratio of 0.3 was used for all mixtures produced. The mixtures considered for this study are optimal mixtures of each phase. The optimal combinations (for Phases I and III) were at 20% RBW and 80% RCT, $SiO_2/Al_2O_{3,}$ and Na_2O/SiO_2 ratios of 10.0 and 0.24, respectively. In Phase III, the highest improvements were obtained by preliminary treatment under 100 °C for 24 h. In the ternary mixtures, incorporating 45%, FC achieved an 88% improvement in compressive strength at 28 days.

Goal and scope

This study aims to carry out an LCA analysis on a previous

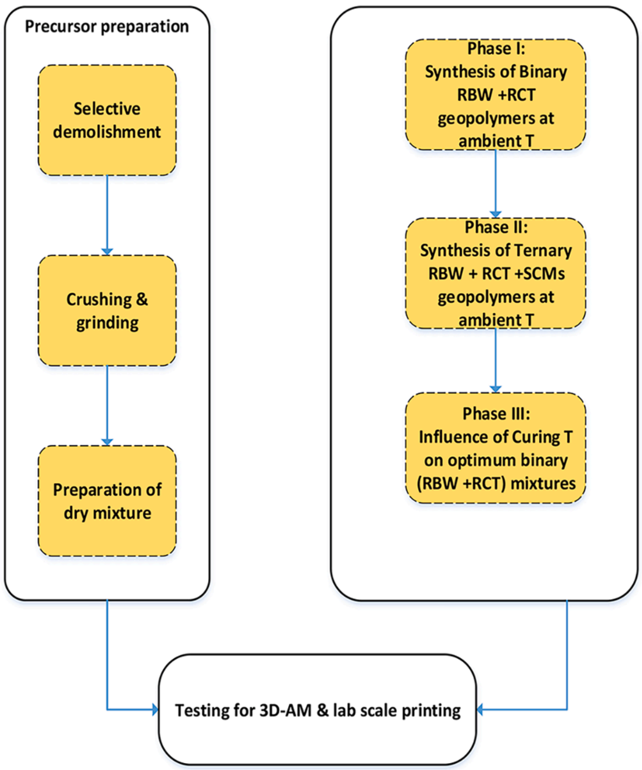

Fig. 1. Flow chart for the preparation and procedure of the experimental geopolymers.

experimental paper (Mahmoodi et al., 2021) to assess the environmental impacts of CDW-based geopolymers. An iterative and non-linear procedure is followed using GaBi software (Sphera - GaBi solutions, 2020) to carry out the LCA analysis. GaBi (Sphera - GaBi solutions, 2020) is one of the leading LCA software model elements in a product or system from an LCA perspective. It provides access to the databases with details of the costs, energy, and environmental impacts of refining raw materials and process components. It can calculate the environmental impact of the studied product or system (Sphera - GaBi solutions, 2020). This study is performed based on the international organisation for standardisation (ISO) guidelines; Fig. 2 shows the framework based on ISO 14040 and 14044 standards. It includes the goal and scope definition, life cycle inventory analysis, life cycle impact assessment, and interpretation.

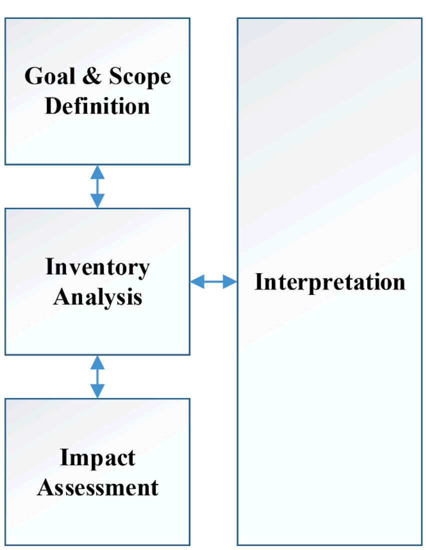

Fig. 2. ISO 14040 and 14044 LCA framework.

Fig. 3 exhibits the LCA boundary diagram for producing geopolymer binders and the plan of the LCA study. The base case for the LCA analysis delivers 1 m^3 of geopolymer. The optimised blend of the precursor materials used is 20% RBW and 80% RCT, SiO_2/Al_2O_3, and Na_2O/SiO_2 ratios of 10.0 and 0.24, respectively, for Phase I (Mahmoodi et al., 2021) and it is cured under ambient temperature. This is taken from the previous paper's experimental section of Phase I. A ternary scenario is also studied in Phase II, where M, GGBS, FC, and FS are included in addition to the RBW and RCT as precursors. The LCA of the best-case scenario for ternary geopolymer binder production (the mixture incorporating 45% FC from Phase II) is also done. Finally, for Phase III, the impacts of curing the mix at 100 °C are found by conducting another LCA analysis. The optimal mixture for Phase III was the same as for Phase I. The midpoint TRACI 2.0 methodology is used for the LCIA, available in GaBi software (Sphera - GaBi solutions, 2020). The results are analysed in the interpretation section, and an industrial-scale case study is conducted for the mixtures of Phases I, II, and III.

Inventory analysis

In this section, the inventory data of the system is collected and summarised. It consists of the energy and material inputs into the system boundary. The authors of the experimental study (Mahmoodi et al., 2021) have provided the inventory data for the analysis. Processes in the GaBi software databases include the life cycle of the process from its raw material extraction to the waste disposal on a case-to-case basis. The mass and energy values are manually entered into the GaBi software, and the LCA model is constructed. The inventory data is calculated based on the final product's functional unit of 1 m3. The total power requirements of the machinery and heating are presented in Table 1. These power requirements entered in GaBi software are assumed to be electrical energy from the grid in Canada. Table 2 shows the transportation details of the CDWs, SCMs, and alkali activators. Diesel is used as fuel for vehicles transporting the materials from the relevant site to the laboratory.

Table 3 shows the mass flows into the process per 1 m^3 of the final product. Phases I, II, and III show different proportions of the CDW and material inputs based on the optimum mixtures from the experiment (Mahmoodi et al., 2021). For Phase I, the information used is 20% RBW and 80% RCT, SS/Al_2O_3, and Na_2O/SS ratios of 10.0 and 0.24, respectively. In Phase II (ternary mixtures), the optimum results are combinations incorporating 45% FC. In Phase III, the optimum mix had the same composition as in Phase I; however, it includes heating for curing the final product.

Table 4 shows the input energy flows of 1 m^3 of the products prepared with the mixtures from Phases I, II, and III into the system. The transportation, precursor preparation, alkali activator preparation, operation, and curing require energy inputs in electricity, fuel, and heat. The precursor preparation needs machinery for crushing, milling, and sieving the CDW before mixing it with the alkaline activators. Electrical energy from the grid is input into GaBi software for these machines. The production flows for the alkaline activator of SH are taken from GaBi databases. The SH is produced using an electrolysis process 'mix' including mercury, diaphragm, and membrane cells (Sphera - GaBi solutions, 2020). SS data is not available in the GaBi databases; therefore, the thermal energy of 35 MJ/kg (Federal Environmental Agency, 2001) from natural gas (Soluble, n.d.) is assumed to make the calculation available in the Canada database. Phase III requires curing at 100 °C, and thermal energy is used to model this process.

Another case study is also modelled for a large-scale production scenario. In this scenario, the precursor preparation and mixing would be done using industrial-sized machines with lower energy input. The LOESCHE mill type LM 45.4 (Strohmeyer, 2015) can perform crushing and grinding operations together. The LM 45.4 mill has an installed drive power of 2000 kW. Production capacity, which may differ

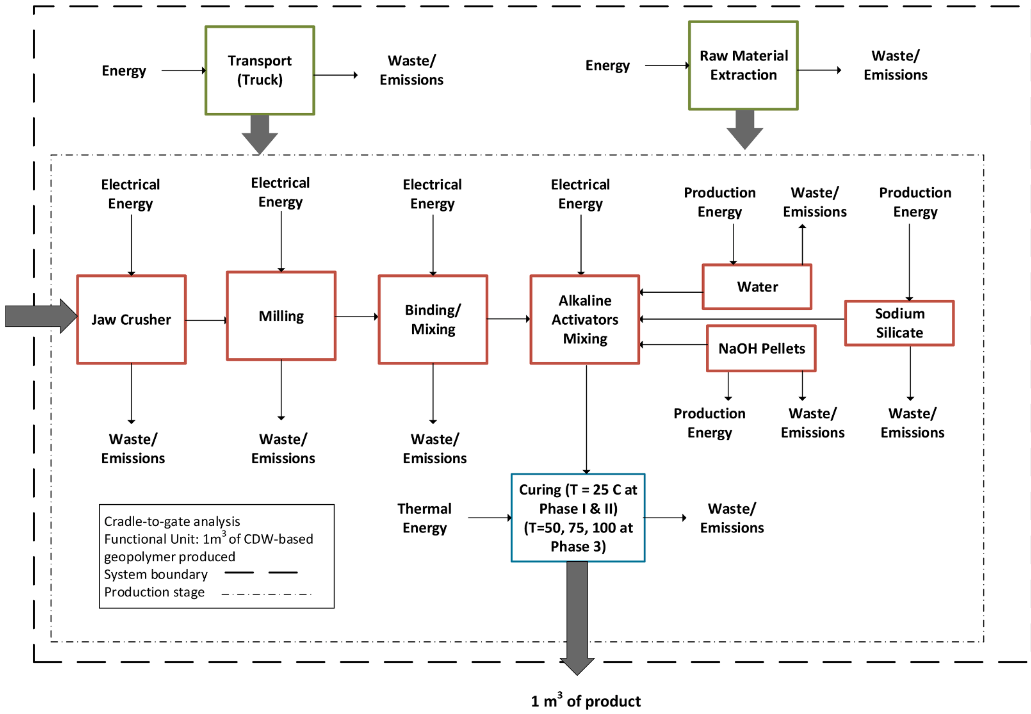

Fig. 3. LCA boundary diagram for producing geopolymer binders.

Table 1
Consumption of power by machinery used.

Process	Power (Watt)
Ball Mill (Electrical)	1500
Jaw Crusher (Electrical)	1500
Sieving (Electrical)	250
Mixing (Electrical)	600
Heat Curing	2200

Table 2
Transportation inventory data for material collection.

Material	Mass transported (ton)	Fuel (L/ 100 km)	Distance (km)	Fuel Energy Consumption (L/ ton)
CDW (Truck)	10	22	20	0.44
Alkali Activators (Pickup truck)	0.5	15	30	9
SCM (Truck)	10	22	10	0.22

according to the material, ranges from approximately 190-to 400 tonnes/hour. Therefore, as an assumption, 200 tonnes/hour is used. In addition to the industrial/large-scale grinding machines, a large-scale industrial mixer can also be used for large-scale mixture preparation. CPB 120 mixer is used (Contec, n.d.) with a capacity of 120 m^3/h and a total installed power of 190 kW. Table 5 shows the energy requirements of Phases I, II, and III mixtures for the industrial-sized scenario of about 1 m^3 of geopolymer.

Life cycle impact assessment

The LCIA results obtained by using TRACI 2.0 on GaBi software are presented in this section. The results are represented in two prominent cases: the base case of the developed process and the improved case based on the environmental analysis performed in the base case.

Base case scenario for Phase I, II, and II:

As the overall process has been divided into three main phases, for each phase, the LCIA is represented in this section. Table 6 shows the environmental impacts of the different mixtures from Phases I, II, and III. Phase III has the highest GWP, AP, EP, ODP, and smog emissions due to the heating required to cure the final product. However, adding curing is the only difference between Phase I and III, and this does not seem to make much of a difference to environmental impacts. Even though the ecological effects in the absence of heat curing are lower, if the geopolymer binder produced has increased performance due to curing, it can be feasible to choose the curing option as the impact difference is not significantly high. Phase II shows the lowest environmental impacts for all the categories studied. Phase II is a ternary

Table 3
Material inputs into the system for 1 m^3 of binary and ternary geopolymer production.

	Mixture Proportion (kg/m^3)								
	RBW* (kg/m^3)	RCT* (kg/m^3)	M* (kg/m^3)	G* (kg/m^3)	FC* (kg/m^3)	FF* (kg/m^3)	SS* (kg/m^3)	SH* (kg/m^3)	Compressive strength (MPa)
Phase I	226.0	904.0	–	–	–	–	249.0	510.0	30
Phase II	117.0	527.0	–	–	527.0	–	340.0	358.0	56.5
Phase III	226.0	904.0	–	–	–	–	249.0	510.0	52.3

*RBW: Recycled brick waste, RCT: Recycled ceramic tile, M: Metakaolin, GGBS: Ground granulated blast furnace slag, FC: Fly Ash C Class, FF: Fly Ash F Class, SS: Na_2SiO_3, SH: NaOH, W: Water.

Table 4
Energy inputs into the system for 1 m^3 of binary and ternary geopolymer production.

	Energy Consumption (MJ/m^3)								
	Transportation		Precursor Preparation			Alkali Activator		Operation	Curing
	CDW/SCM	Alkali Activator	Jaw Crusher	Milling	Sieving	SS	SH	Mixing	
Phase I	19.2	264.4	145.3	1743.4	16.95	8715	10,455	18	–
Phase II	15.5	243.1	82.8	993.6	9.7	11,900	7339	18	–
Phase III	19.2	264.4	145.3	1743.4	16.95	8715	10,455	18	178

Table 5
Energy inputs for a large-scale model to produce 1 m^3 of binary and ternary geopolymer.

	Energy Consumption (MJ/m^3)						
	Transportation		Precursor Preparation	Alkali Activator		Operation	Curing
	CDW/SCM	Alkali Activator	Crushing and Grinding	SS	SH	Mixing	
Phase I	19.24164	264.3597	40.68	8715	10,455	5.7	–
Phase II	15.5	243.1134	23.184	11,900	7339	5.7	–
Phase III	19.24164	264.3597	40.68	8715	10,455	5.7	178

Table 6
LCIA results were obtained by using TRACI 2.0 methodology on GaBi Software to produce 1 m^3 of geopolymer.

	GWP (kg CO_2-eq)	AP (kg SO_2-eq)	EP (kg N-eq)	ODP (kg CFC 11-eq)	Smog (kg O_3-eq)
Phase I	1800.98	4.87	0.26	7.49E-10	77.48
Phase II	1631.90	3.68	0.20	5.14E-10	66.48
Phase III	1811.14	4.87	0.26	7.49E-10	77.71

mixture of RBW + RCT + SCM, giving better ecological results than Phases I and III, binary mixtures of RBW and RCT.

Phase I: The distribution of input inventory variables for impact categories in Phase I (geopolymer production with the binary blend of RBW and RCT) is shown in Fig. 4. The graph shows the most significant flows affecting the environmental impacts. For GWP, SS and SH production has the highest effect. SH has the highest impact on the AP, EP, ODP, and smog. Transportation does not cause significant environmental impacts in these categories. SS overall poses less harm than others to the ecological effects studied.

Phase II has lower environmental impacts than the other phases; the detailed analysis is shown in Fig. 5.

It shows that the impacts of SS on the GWP, AP, EP, and smog increase in percentage. For example, for GWP, the effects of SS increased from around 35% to 54%. This is because SCMs (45% FC) were incorporated into the mixture in Phase II, resulting in a different mixture composition. Phase II has lower environmental impacts than the other phases for all categories studied.

Phase III: The inventory compositions for Phase III are the same as those for Phase I. However, there is an additional curing stage where the mixture is heated to 100 °C. This step improves the physical characteristics of the geopolymer product; however, it also increases the environmental impact due to extra heating.

Fig. 6 shows the distributions of the ecological impact results of the mixture from Phase III. The distributions of the environmental effects do not change significantly compared to Phase I. The curing causes a slight increase in GWP and smog as it was assumed to be thermal energy in the software. Thermal energy from natural gas is derived from fossil fuels resulting in CO_2 and other greenhouse gas emissions.

Improved case - Large-scale scenario for Phases I, II, and III

Since the results showed that the electrical energy needed for material processing contributes significantly to the overall environmental impact, one of the critical aspects of this could be the smaller capacity of the lab-scale machinery for crushing, grinding, and mixing processes.

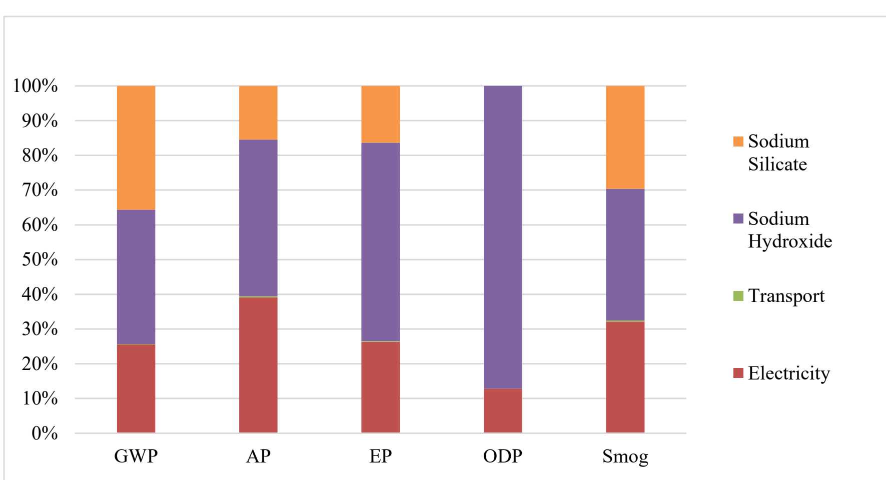

Fig. 4. Impact assessment results of Phase I for 1 m^3 of geopolymer binder produced.

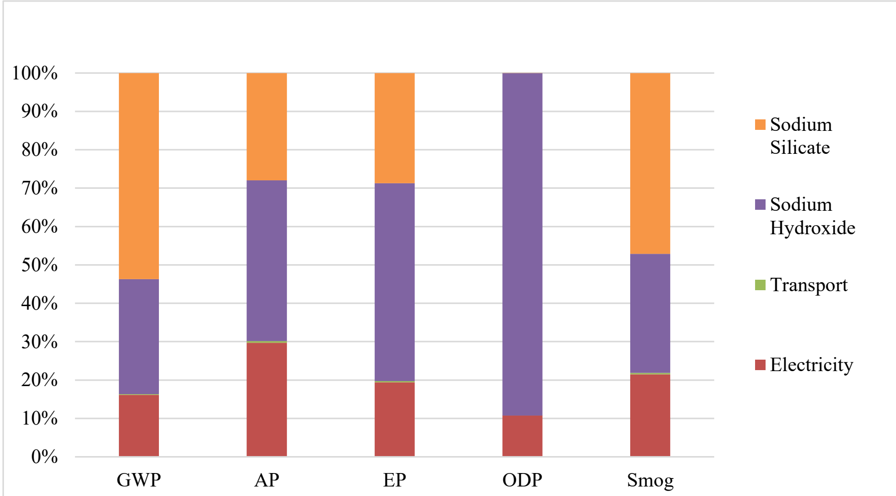

Fig. 5. Distribution of impact assessment results of Phase II for 1 m^3 of geopolymer binder produced.

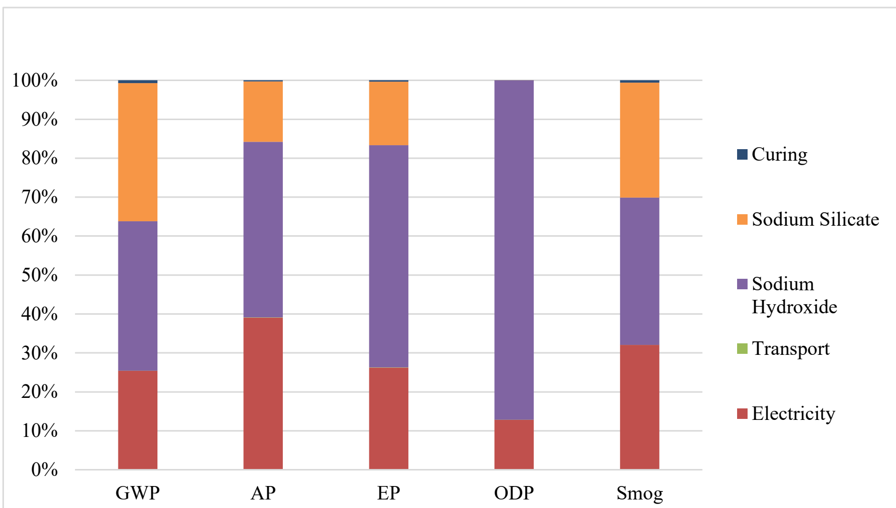

Fig. 6. Distribution of impact assessment results of Phase III for 1 m^3 of geopolymer binder produced.

Hence, in this scenario, Phases I, II, and III mixtures are modelled based on a large industrial scale. The machinery used for grinding, milling, and mixing are replaced by industrial-scale with high loading capacity and lower energy inputs per m^3.

Table 7 shows the large-scale case's LCIA results, utilising the LOESCHE mill type LM 45.4 (Strohmeyer, 2015) for crushing and grinding and the CPB 120 for mixing (Contec, n.d.). Fig. 7 compares the lab-scale scenario in Table 5 and the industrial-scale method in Table 7. The graph is calculated by normalising the data to the maximum values in the data set.

Most environmental impacts are reduced by utilising industrial-scale machinery as the electrical input was less overall. For GWP, the lowest large-scale scenario is Phase I, followed by Phases III and II. The most everyday large-scale GWP value of 1352 kg CO_2-eq is 448.86 kg CO_2 eq less than the lab-scale Phase I value. Furthermore, this value is around 25% less than the highest GWP value in laboratory-small and industrial-large scale scenarios. Moreover, for AP, the large-scale Phase II value of 2.61 is 46% lower than the highest AP value in the graph. For EP, incorporating industrial machinery in significant scale phase II analysis has reduced the impacts by 37% compared to the highest value. The ODP is also the lowest in the large-scale phase II analysis. The smog impact is decreased using industrial-scale machinery, and Phases I, II, and III for large scale show similar values around 53 kg O_3 eq.

Salas et al. (2018) found that the GWP of geopolymer concrete ranged between 84 kg CO_2 eq/m^3 and 254 kg CO_2 eq/m^3 for various compressive strengths. The values found in this study are significantly higher because it is a cradle-to-gate analysis and considers the raw material extraction and all of the relevant processes for material and energy flows such as SH, SS, thermal energy, electricity, etc. Moreover, the location of the LCA can affect the LCA results as the electricity mix is different for each country, and other methodologies and transportation distances are used.

Table 7
Impact assessment results for phases I, II, and III on a large industrial scale.

	GWP (kg CO_2-eq)	AP (kg SO_2-eq)	EP (kg N-eq)	ODP (kg CFC 11 eq.)	Smog (kg O_3 eq)
Large scale I	1352.12	3.01	0.19	6.56E-10	53.18
Large scale II	1374.82	2.61	0.16	4.60E-10	52.56
Large scale III	1365.23	3.03	0.19	6.56E-10	53.65

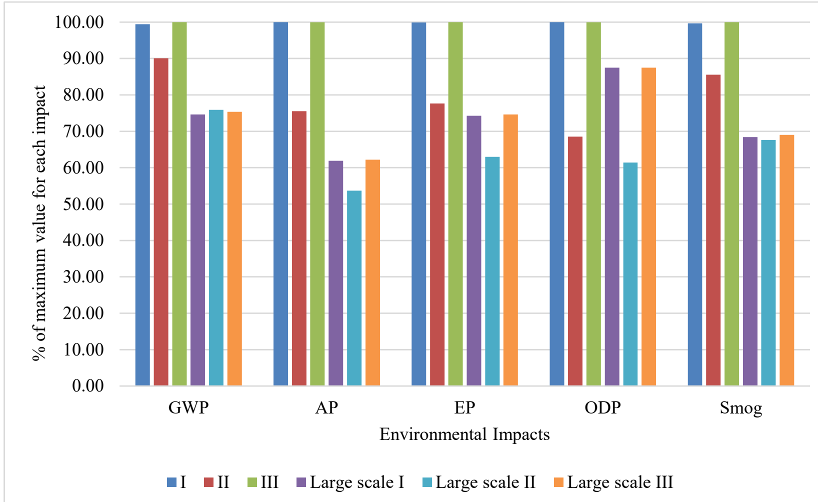

Fig. 7. Comparative graph showing percentage differences between lab-scale and large-scale Phases I, II, and III scenarios.

Conclusion

This study conducts an LCA analysis on an experimental investigation to produce RBW + RCT-based geopolymers. The cradle-to-gate LCA analysis is done using GaBi software following ISO 14040 standards. The inventory data includes the material and energy flow to create 1 m^3 of geopolymer binder. The LCA analysis is performed on three different mixtures in terms of composition or curing (Phase I, II, and III), which were found to be optimum in terms of physical properties from the experimental paper. For Phase I, a binary composition mixture was studied with 20% RBW, 80% RCT, SiO_2/Al_2O_3 ratio of 10, and Na_2O/SiO_2 ratio of 0.24. For Phase II, a ternary mixture was studied, including an addition of 45% FC to the RCB + RCT. In Phase III, the optimum mix had the same mixture composition as in Phase I; however, it includes heating for curing the final product. The LCIA used TRACI 2.0 methodology and found the GWP, AP, EP, ODP, and impact categories. For Phases I, II, and III, the GWP values were 1801, 1632, and 1811 kg CO_2 eq.

- In Phase I, SH has a high contribution to the GWP, AP, EP, ODP, and smog environmental impacts, followed by the electricity and sodium silicate production;
- For the ternary composition of Phase II, there is a slight increase in the contribution of SS to the overall impact categories. This composition has the lowest environmental impacts; therefore, the use of FC in ternary geopolymers is more environmentally beneficial than the binary compositions in this case;
- Phase III shows similar results to Phase I but has 11 kg CO_2 eq higher GWP due to added curing;
- For the sizeable industrial case scenario, less electrical input was used by the machinery overall; therefore, it reduced the environmental impacts significantly. At a large scale, phase I had the lowest GWP of 1352 kg CO_2 eq which was 25% less than the highest GWP value.

Future recommendations

Performing LCA on different geopolymer binders with varying compositions and mechanical properties will lead to the development of geopolymers by considering their environmental effects at the design stage. This will ensure that the concept of sustainability is adapted to the design phase, and therefore the sustainability performance of the final product will be improved. Moreover, such studies' social and economic impacts can also be addressed early. Additionally, there should be more sustainable methodologies to produce sodium hydroxide and sodium silicate, which can be incorporated into geopolymer production to reduce environmental impacts significantly.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Alhaj, M., Tahir, F., Al-Ghamdi, S.G., 2022. Life-cycle environmental assessment of solar-driven Multi-Effect Desalination (MED) plant. Desalination 524, 115451. https://doi.org/10.1016/j.desal.2021.115451.

Andrew, R.M., 2018. Global CO2 emissions from cement production. Earth Syst. Sci. Data 10, 195–217. https://doi.org/10.5194/ESSD-10-195-2018.

Bajpai, R., Choudhary, K., Srivastava, A., Sangwan, K.S., Singh, M., 2020. Environmental impact assessment of fly ash and silica fume based geopolymer concrete. J. Cleaner Prod. 254, 120147 https://doi.org/10.1016/j.jclepro.2020.120147.

Colangelo, F., Farina, I., Travaglioni, M., Salzano, C., Cioffi, R., Petrillo, A., 2021. Eco-efficient industrial waste recycling for the manufacturing of fibre reinforced innovative geopolymer mortars: Integrated waste management and green product development through LCA. J. Cleaner Prod. 312, 127777 https://doi.org/10.1016/J.JCLEPRO.2021.127777.

CONTEC. Concrete batching plants and concrete mixers. n.d.

Dadsetan, S., Siad, H., Lachemi, M., Sahmaran, M., 2019. Construction and demolition waste in geopolymer concrete technology: A review. Mag. Concr. Res. 71, 1232–1252. https://doi.org/10.1680/JMACR.18.00307.

Duxson, P., Fernández-Jiménez, A., Provis, J.L., Lukey, G.C., Palomo, A., van Deventer, J.S.J., 2007. Geopolymer technology: the current state of the art. J. Mater. Sci. 42 (9), 2917–2933.

Federal Environmental Agency. Large Volume Solid Inorganic Chemicals: Sodium Silicate 2001.

Ghadir, P., Zamanian, M., Mahbubi-Motlagh, N., Saberian, M., Li, J., Ranjbar, N., 2021. Shear strength and life cycle assessment of volcanic ash-based geopolymer and cement stabilized soil: A comparative study. Transp. Geotech. 31, 100639 https://doi.org/10.1016/J.TRGEO.2021.100639.

Guo, X., Shi, H., Chen, L., Dick, W.A., 2010. Alkali-activated complex binders from class C fly ash and Ca-containing admixtures. J. Hazard. Mater. 173, 480–486. https://doi.org/10.1016/J.JHAZMAT.2009.08.110.

Hamdi, N., Ben Messaoud, I., Srasra, E., 2019. Production of geopolymer binders using clay minerals and industrial wastes. C. R. Chim. 22, 220–226. https://doi.org/10.1016/J.CRCI.2018.11.010.

Huynh, T.-P., Pham, V.-H., Do, N.-D., Nguyen, T.-C., Ho, N.-T., 2021. Performance evaluation of pre-foamed ultra-lightweight composites incorporating various proportions of slag. Periodica Polytechnica Civ. Eng. 65, 276–286. https://doi.org/10.3311/PPCI.16996.

S.A. Khan, N. Mir, A. Kul, O. Şahin, M. Şahmaran, M. Koç. The 5th International Conference on Electrical Engineering and Green Energy, CEEGE 2022, 8-11 June, Berlin, Germany. Renewable Energy for Carbon Footprint Reduction and Optimization of Green Geopolymers Material's Production, Berlin, Germany: 2022.

Khan, S.A., Koç, M., Al-Ghamdi, S.G., 2021. Sustainability assessment, potentials and challenges of 3D printed concrete structures: A systematic review for built environmental applications. J. Cleaner Prod. 303, 127027 https://doi.org/10.1016/j.jclepro.2021.127027.

Liu, Q., Tong, T., Liu, S., Yang, D., Yu, Q., 2014. Investigation of using hybrid recycled powder from demolished concrete solids and clay bricks as a pozzolanic supplement for cement. Constr. Build. Mater. 73, 754–763. https://doi.org/10.1016/J.CONBUILDMAT.2014.09.066.

Luhar, I., Luhar, S., Abdullah, M.M.A.B., Nabiałek, M., Sandu, A.V., Szmidla, J., Jurczyńska, A., Razak, R.A., Aziz, I.H.A., Jamil, N.H., Deraman, L.M., 2021. Assessment of the suitability of ceramic waste in geopolymer composites: an appraisal. Materials 14 (12), 3279.

Mahmoodi, O., Siad, H., Lachemi, M., Dadsetan, S., Sahmaran, M., 2021. Development and characterization of binary recycled ceramic tile and brick wastes-based geopolymers at ambient and high temperatures. Constr. Build. Mater. 301, 124138 https://doi.org/10.1016/j.conbuildmat.2021.124138.

Monteiro, P.J.M., Miller, S.A., Horvath, A., 2017. Towards sustainable concrete. Nature Mater 16 (7), 698–699. https://doi.org/10.1038/nmat4930.

Ouellet-Plamondon, C., Habert, G., 2015. Life cycle assessment (LCA) of alkali-activated cements and concretes. In: Handbook of Alkali-Activated Cements Mortars and Concretes. https://doi.org/10.1533/9781782422884.5.663.

Provis, J.L., 2018. Alkali-activated materials. Cem. Concr. Res. 114, 40–48. https://doi.org/10.1016/J.CEMCONRES.2017.02.009.

Rebitzer, G., Ekvall, T., Frischknecht, R., Hunkeler, D., Norris, G., Rydberg, T., Schmidt, W.-P., Suh, S., Weidema, B.P., Pennington, D.W., 2004. Life cycle assessment. Environ. Int. 30 (5), 701–720. https://doi.org/10.1016/j.envint.2003.11.005.

Salas, D.A., Ramirez, A.D., Ulloa, N., Baykara, H., Boero, A.J., 2018. Life cycle assessment of geopolymer concrete. Constr. Build. Mater. 190, 170–177. https://doi.org/10.1016/j.conbuildmat.2018.09.123.

Shi, C., Qu, B., Provis, J.L., 2019. Recent progress in low-carbon binders. Cem. Concr. Res. 122, 227–250. https://doi.org/10.1016/J.CEMCONRES.2019.05.009.

Shi, X., Zhang, C., Liang, Y., Luo, J., Wang, X., Feng, Y., et al., 2021. Life Cycle Assessment and Impact Correlation Analysis of Fly Ash Geopolymer Concrete. Materials 14, 7375. https://doi.org/10.3390/MA14237375.

Silva, G., Castañeda, D., Kim, S., Castañeda, A., Bertolotti, B., Ortega-San-Martin, L., Nakamatsu, J., Aguilar, R., 2019. Analysis of the production conditions of geopolymer matrices from natural pozzolana and fired clay brick wastes. Constr. Build. Mater. 215, 633–643. https://doi.org/10.1016/J.CONBUILDMAT.2019.04.247.

SOLUBLE SODIUM SILICATE MANUFACTURE n.d.

Sphera - GaBi solutions. SPHERA n.d. https://gabi.sphera.com/international/index/ (accessed April 15, 2020).

Strohmeyer, D., 2015. Latest technological innovations in grinding with the vertical roller mill. vol. 13.

Ulugöl, H., Kul, A., Yıldırım, G., Şahmaran, M., Aldemir, A., Figueira, D., et al., 2021. Mechanical and microstructural characterization of geopolymers from assorted construction and demolition waste-based masonry and glass. J. Cleaner Prod. 280, 124358. https://doi.org/10.1016/J.JCLEPRO.2020.124358.

United Nations Department of Economic and Social Affairs Population Division. 68% of the world population projected to live in urban areas by 2050, says UN | UN DESA | United Nations Department of Economic and Social Affairs. United Nations News 2018.

Xu, F., Gu, G., Zhang, W., Wang, H., Huang, X., Zhu, J., 2018. Pore structure analysis and properties evaluations of fly ash-based geopolymer foams by chemical foaming method. Ceram. Int. 44, 19989–19997. https://doi.org/10.1016/j.ceramint.2018.07.267.

Zanelli, C., Conte, S., Molinari, C., Soldati, R., Dondi, M., 2021. Waste recycling in ceramic tiles: a technological outlook. Resour. Conserv. Recycl. 168, 105289 https://doi.org/10.1016/j.resconrec.2020.105289.